

Healthcare Drug Effectiveness Analysis Using Statistical Testing and Machine Learning

1. Project Overview

Project Title

Healthcare Drug Effectiveness Analysis and Patient Recovery Prediction Using Statistical Testing and Machine Learning

Domain

Healthcare Analytics

Objective

The objective of this project is to evaluate the effectiveness of three different drugs (Drug A, Drug B, and Drug C) using statistical hypothesis testing and machine learning techniques. The project aims to identify which treatment provides better patient outcomes, understand the factors affecting recovery, and build predictive models to estimate whether a patient is likely to recover.

2. Business Problem Statement

A hospital is conducting a clinical study to compare three different drugs used for treating patients with high blood pressure.

The management wants to answer the following questions:

- Which drug reduces blood pressure the most?
- Which drug provides the fastest recovery?
- Does age influence recovery?
- Does gender influence recovery?
- Does pain score affect recovery?
- Which treatment should doctors recommend?
- Can we predict whether a future patient will recover?

The answers to these questions help doctors make evidence-based treatment decisions and improve patient outcomes.

3. Dataset Description

The dataset contains information about patients who received one of three treatments.

Feature	Description
Age	Patient's age
Gender	Male/Female
Treatment	Drug A, B or C
BP_Before	Blood pressure before treatment
BP_After	Blood pressure after treatment
BP_Reduction	Reduction in blood pressure
RecoveryDays	Number of days required to recover
PainScore	Pain level after treatment
Recovered	Target variable (Yes/No)

4. Project Workflow

1. Business Understanding
2. Data Collection
3. Data Cleaning
4. Feature Engineering
5. Exploratory Data Analysis (EDA)
6. Statistical Hypothesis Testing
7. Machine Learning Model Development
8. Model Evaluation
9. Feature Importance Analysis
10. Business Recommendations

5. Exploratory Data Analysis (EDA)

EDA was performed to understand the characteristics of patients and identify relationships between variables before building statistical and machine learning models.

The following analyses were performed:

- Age distribution

- Gender distribution
- Treatment distribution
- Recovery distribution
- Blood pressure before treatment
- Blood pressure after treatment
- Blood pressure reduction
- Recovery days
- Pain score analysis
- Correlation analysis
- Treatment vs Recovery
- Treatment vs Blood Pressure Reduction
- Treatment vs Recovery Days
- Gender vs Recovery
- Age vs Recovery

The objective of EDA was to discover hidden patterns and understand the data before performing statistical analysis.

6. Statistical Hypothesis Testing

Statistical testing was performed to determine whether the observed differences in the data were statistically significant or occurred due to random chance.

6.1 ANOVA Test

Why was ANOVA used?

ANOVA compares the average blood pressure reduction among more than two treatment groups.

Research Question

Do all three drugs reduce blood pressure equally?

Null Hypothesis (H_0)

All treatments have the same average blood pressure reduction.

Alternative Hypothesis (H_1)

At least one treatment has a different average blood pressure reduction.

Business Interpretation

If the p-value is less than 0.05, the treatments are not equally effective.

6.2 Independent T-Test

Why was the T-Test used?

To compare the average blood pressure reduction between male and female patients.

Research Question

Does gender influence blood pressure reduction?

Null Hypothesis

Male and female patients have the same average blood pressure reduction.

Business Interpretation

This test determines whether treatment effectiveness differs between genders.

6.3 Z-Test

Why was the Z-Test used?

To compare the average blood pressure reduction between two treatment groups when the sample size is sufficiently large.

Business Question

Is Drug B significantly different from Drug C?

Business Interpretation

This helps identify whether one treatment performs better than another.

6.4 Mann-Whitney U Test

Why was this test used?

Recovery days may not follow a normal distribution.

The Mann-Whitney U Test compares recovery time between two independent groups without assuming normality.

Business Question

Do male and female patients recover in the same number of days?

6.5 Chi-Square Test

Why was Chi-Square used?

Recovery status and treatment type are categorical variables.

The Chi-Square Test determines whether treatment and recovery are associated.

Business Question

Does the treatment influence patient recovery?

Business Interpretation

If the p-value is less than 0.05, treatment selection has a significant impact on recovery.

7. Machine Learning

The objective of machine learning was to predict whether a patient will recover based on patient characteristics and treatment information.

Target Variable

Recovered (Yes/No)

Logistic Regression

Why was Logistic Regression selected?

- Baseline classification model
- Easy to interpret
- Widely used in healthcare
- Produces probability estimates

Purpose

Predict the probability of patient recovery.

Decision Tree

Why was Decision Tree selected?

Decision Trees learn rule-based decisions similar to clinical decision-making.

Purpose

Understand how the model makes decisions and identify the most important variables.

Random Forest

Why was Random Forest selected?

Random Forest combines multiple Decision Trees to improve prediction accuracy and reduce overfitting.

Purpose

Build a more robust model using ensemble learning.

XGBoost

Why was XGBoost selected?

XGBoost is an advanced boosting algorithm that sequentially improves prediction errors made by previous trees.

Purpose

Evaluate whether boosting techniques outperform traditional machine learning algorithms.

8. Model Evaluation

The following evaluation metrics were used:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- Feature Importance

These metrics provide a comprehensive assessment of model performance beyond simple accuracy.

9. Model Comparison

Model	Purpose
Logistic Regression	Baseline predictive model
Decision Tree	Rule-based prediction and interpretation
Random Forest	Ensemble learning with multiple trees
XGBoost	Gradient boosting for higher predictive performance

The best-performing model should be selected based on predictive performance and interpretability.

10. Feature Importance

Feature importance analysis identifies which variables contribute most to predicting patient recovery.

Typical important variables include:

- Treatment
- Blood Pressure Reduction
- Age
- Blood Pressure After Treatment

This analysis helps clinicians understand which factors influence recovery the most.

11. Business Insights

The project provides several actionable insights:

- Treatment effectiveness differs across drugs.
 - Blood pressure reduction is an important indicator of recovery.
 - Patient age influences recovery probability.
 - Pain score has a negative relationship with recovery.
 - Gender has relatively little influence on recovery.
 - Statistical testing confirms whether treatment differences are significant.
 - Machine learning enables prediction of recovery for future patients.
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12. Conclusion

This project combines statistical analysis and machine learning to evaluate drug effectiveness in a healthcare setting.

Statistical hypothesis testing validates whether differences between treatments are statistically significant, while machine learning predicts future patient outcomes.

The project demonstrates an end-to-end healthcare analytics workflow, including business understanding, exploratory data analysis, hypothesis testing, predictive modeling, feature importance analysis, and business recommendations.

The resulting workflow supports evidence-based treatment selection and provides a foundation for future clinical decision support systems.